K-means Homework: Can We Justify These Groups?

Printable companion; see the Markdown source for clickable links.

Format: Individual portfolio investigation.

Estimated effort: 4-6 hours plus independent peer validation.

Purpose and scope

Demonstrate the full path from measurements to centroids and from cluster scores to a defensible interpretation. Use the guide, lab, and notebook as cited references, not as a completed submission. No customer data collection or 85-million-row execution is required.

Part 1: Hand calculations before code

These are four invented customers. Inputs are monthly orders and average items per order, not known customer segment labels.

Customer	Monthly orders	Average items per order
H	1	1
I	3	2
J	8	6
L	10	8

Use k = 2, initial center 0 = H (1, 1), and center 1 = J (8, 6). Use raw features for this hand exercise only. An equal-distance tie goes to the smaller center index. Do not round until reporting four decimal places.

1. Explain why this is unsupervised and how k differs from KNN's k.

2. Calculate each customer's squared Euclidean distance to both initial centers. Show all substitutions for I. Assign all four customers.

3. Calculate each updated centroid by averaging each assigned feature. Repeat assignment/update until centers do not change. Show a table for every iteration, not only the final answer.

4. Calculate total squared error before and after the first update, then at stopping. Explain why averaging improves the objective for fixed assignments.

5. Assign new customer T = (3, 1) using the final centers, without refitting. Does assigning T update a center? Is the distance a membership probability?

K-means homework

Continued

6. Can reaching max_iter alone prove convergence? Explain why different initializations may produce different results and why cluster 0 is not a known semantic class.

Calculate a separate scaling example from training rows (2, 10), (2, 50), (8, 10), (8, 50), where feature 1 is monthly orders and feature 2 is dollars. Compute both training means and population standard deviations by hand (divide squared deviations by 4, not 3), then scale query (8, 50). Explain why these statistics prepare the geometry but do not replace assigning records to centroids. Do not mix these rows with H, I, J, L.

Part 2: Reproducible coding investigation

Use the supplied deterministic make_customers(seed=42) generator: 600 synthetic records, two named features, no target. You may reuse/cite the generator and lab helpers, but write your own documented end-to-end script and explain your decisions.

1. Report features/units, row count, missingness, and descriptive ranges. Explain that the simulation is invented and deliberately easy.

2. Use the lab's exact two-stage seed-42 index split: 20% test, then 25% of development for validation, preserving returned order. Verify disjointness and coverage. State why labels are not available to stratify.

3. Fit StandardScaler on training only. Fit full KMeans for k in [2, 3, 4, 5, 6], using init="k-means++", n_init=10, max_iter=300, tol=1e-4, random_state=42, algorithm="lloyd".

4. For each k, report training mean squared distance, validation mean squared distance, validation silhouette, and n_iter_. Plot compactness and silhouette against k. Select highest validation silhouette; exact score ties choose smaller k. Do not select by smallest training inertia.

5. For the selected k, fit five training-only, single-initialization models with seeds 0-4 and n_init=1. Compare their predictions on the same validation rows using pairwise adjusted Rand index. Explain what this stability check does and does not establish; it has no true customer labels.

6. Freeze k and preprocessing choices. Refit a new scaler and full KMeans on the original 480 development indices, then score the 120 test records once.

7. Fit MiniBatchKMeans using the same fixed development scaler and k, with batch_size=64, n_init=1, random_state=42, and five caller-controlled shuffled passes of partial_fit, as in the lab. Compare test mean squared distance and silhouette without using test results to retune anything.

8. Explain why five passes are a computational budget, not proof of convergence, and why partial_fit does not enforce a global max_iter loop. Full KMeans n_init means full restarts; MiniBatchKMeans initialization selection is not the same as ten complete mini-batch training runs.

K-means homework

Continued

9. Inverse-transform full-model centroids into orders/month and dollars, report development cluster sizes, and propose cautious descriptive names. Cluster IDs are arbitrary: do not compare the two models by raw ID equality.

Required files

Your script must regenerate these outputs relative to its own location. Use decimal metrics with at least six places and integer counts/settings.

File	Required content
reports/k_selection.csv	Five rows: k, train_mean_squared_distance, validation_mean_squared_distance, validation_silhouette, n_iter
reports/test_metrics.csv	Two rows: model, mean_squared_distance, silhouette; model values full_kmeans and streamed_minibatch
reports/cluster_profiles.csv	One row per final full-model center: cluster, development_count, orders_per_month, average_basket_dollars
reports/k_selection.png	Both compactness and silhouette curves, labeled
reports/test_clusters.png	Both test assignments plotted in original units with development centers

Also print the chosen k, five-seed pairwise ARI summary, and full-model n_iter_. No particular k or score earns credit by itself. Explain results you actually observe, including equal scores if they occur.

Part 3: Interpretation and scale

1. Interpret your test mean squared distance: what is being averaged and in what coordinate system? Why is it not accuracy? Why does total inertia depend on dataset size and typically decrease with more centers?

2. For one hypothetical record with a = 0.4 and b = 1.0, calculate its silhouette (b-a)/max(a, b). What do a and b represent? Interpret scores near 1, near 0, and below 0. Are these ground-truth correctness probabilities?

3. Explain why stable compact groups could still be useless or harmful, and why outliers, unequal densities, curved groups, and categorical codes can mislead K-means. Suggest a different method for one failure case.

4. Hypothetically, 85 million rows, two float64 features, k=3, 20 full passes: calculate raw-feature memory and point-to-center comparisons per pass and across 20 passes. Then account for ten complete KMeans restarts. Distinguish comparisons from individual coordinate arithmetic operations.

K-means homework

Continued

5. With 100, 000 rows per chunk, how many chunks make one 85-million-row pass? Contrast full Lloyd accumulation (fixed centers during a pass; global sums and counts; update after the pass) with mini-batch updates after each chunk.

6. Describe two-pass global scaler fitting/transformation, representative initialization, a bounded streaming budget plus a development-only stopping check, and how to sample silhouette instead of constructing a huge pairwise distance matrix. K-means compares to k centers, not all other records.

7. Is a train/validation/test split compulsory for purely descriptive clustering? Explain why this lab uses one for future assignment assessment. If customers repeat or deployment is in the future, what must change?

8. Explain how to assign a new customer without retraining and how to monitor drift, changing cluster sizes, distances, and downstream usefulness.

Report and professional portfolio

Submit a report of approximately 700-1, 000 words excluding hand calculations, tables, and figures. Address every interpretation question; include Part 1's complete hand work and show how your chosen k follows the declared rule. Include a short research-process section outside the word count: real challenges/design decisions, investigation, corrections, evidence, and sources (including course code and any assistance). Do not invent problems.

Use your own professional Git repository with a descriptive title such as "Retail Patterns: An Unsupervised Customer Segmentation Study". Include your script, dependency file, report, required outputs, meaningful commits, and an appropriate .gitignore. Do not commit environments, credentials, private customer records, or signed validation forms.

Your root README must be self-contained for a beginner on both Windows and macOS: Python/Git installation links and verification; opening a terminal; actual clone URL and directory commands; separate OS-appropriate environment, dependency, and run commands; a direct interpreter option if PowerShell activation is blocked; tested versions; expected outputs, approximate runtime, and how to open them; and troubleshooting. Do not assume VS Code or depend on the author's absolute paths or private help.

Independent peer validation and Canvas submission

Follow the existing course replication form and its policy. A classmate must obtain a fresh copy of the final repository, follow only your README on their own computer without your help, and verify that all outputs regenerate. Record the actual OS, Python version, branch, and final successful commit. Record/fix failures and repeat independent validation before both parties sign. Do not claim untested platforms.

K-means homework

Continued

Submit an accessible Git URL with branch and validated commit, the signed verification PDF privately on Canvas, and a note identifying optional extensions or required-scope-only completion. Keep the form and private reviewer details out of Git. A ZIP or screenshot is not a substitute. Extra tuning must remain separate and must not use the test results.

Check the student rubric before submitting.